Pandas Practice Challenge — Code Cheat Sheet

1. Load the Dataset

```
import pandas as pd
import numpy as np
df = pd.read_csv("customers.csv")
```

2. Inspect the Data

```
df.head()
df.info()
df.describe()
df.shape
df.columns
```

3. Check Missing Values

```
df.isnull().sum()
```

Remove rows with missing values: `df = df.dropna()`

Fill missing numerical values: `df["Age"] = df["Age"].fillna(df["Age"].median())`

Fill missing text values: `df["City"] = df["City"].fillna("Unknown")`

4. Find & Remove Duplicates

Count duplicates: `df.duplicated().sum()`

View duplicates: `df[df.duplicated()]`

Remove duplicates: `df = df.drop_duplicates()`

5. Rename Columns

```
df = df.rename(columns={
   "CustomerID": "customer_id",
   "FirstName": "first_name",
   "LastName": "last_name"
})
```

Check:

```
df.columns
```

6. Convert Data Types

To numeric: df["Age"] = pd.to_numeric(df["Age"], errors="coerce")

To date: df["JoinDate"] = pd.to_datetime(df["JoinDate"])

Check types: df.dtypes

7. Create a New Column

Combine columns: df["FullName"] = df["FirstName"] + " " + df["LastName"]

Calculate a new value: df["Income_Thousands"] = df["Income"] / 1000

8. Transform Data

Uppercase: df["City"] = df["City"].str.upper()

Lowercase: df["City"] = df["City"].str.lower()

Remove extra spaces: df["City"] = df["City"].str.strip()

9. Merge Datasets

Load the second dataset:

```
orders = pd.read_csv("orders.csv")
```

Merge using the common key:

```
merged_df = pd.merge(
  df,
  orders,
  on="customer_id",
  how="inner"
)
```

Check:

```
merged_df.head()
```

🔎 Quick Reference

Task	Code
Load CSV	pd.read_csv()
First rows	df.head()

Task	Code
Data information	df.info()
Statistics	df.describe()
Shape	df.shape
Missing values	df.isnull().sum()
Remove missing	df.dropna()
Fill missing	df.fillna()
Find duplicates	df.duplicated()
Remove duplicates	df.drop_duplicates()
Rename	df.rename()
Convert numeric	pd.to_numeric()
Convert date	pd.to_datetime()
New column	df["new"] = ...
Merge	pd.merge()